Adwait V. Datar

Postdoctoral Researcher

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Research profile

I work at the intersection of information geometry, learning and optimization, and control theory. My primary research concerns natural-gradient optimization and Wasserstein-geometric approaches to divergences used in machine learning. I also investigate how policy architectures affect the performance and robustness of embodied learning agents.

Experience

Postdoctoral Researcher Jul 2023-Present
Institute for Data Science Foundations, TU Hamburg *Hamburg, Germany*

- Develop theory at the interface of information geometry and learning, including natural-gradient optimization and Wasserstein-geometric regularization for learning and control.
- Co-author of work published at ICML 2026 and in *JMLR*, *TMLR*, and *SIAM Journal on Scientific Computing*; teach and mentor across machine learning, information geometry, and control.

Doctoral Researcher + Postdoctoral Researcher (8 months) Nov 2016-Jun 2023
Institute of Control Systems, TU Hamburg *Hamburg, Germany*

- Developed modular methods for distributed control of heterogeneous multi-vehicle systems, using robust-control tools and integral quadratic constraints to analyze uncertain cooperative dynamics; doctorate defended in November 2022.
- Built an indoor multi-quadrotor laboratory and led source-seeking experiments that connected theoretical guarantees with hardware demonstrations.
- Contributed technical material and preliminary results to the successful Phase 2 proposal for DFG Priority Programme 1914 on cyber-physical networking.

Research Engineer Jul 2012-Aug 2014
Mercedes-Benz R&D India / Daimler R&D *Pune, India / Stuttgart, Germany*

Education

Dr. rer. nat. (PhD) 2022
TU Hamburg *Hamburg, Germany*
Distributed Control of Multi-Vehicle Systems Using Modular Architectures.

MSc, Mathematical Modelling in Engineering (MathMods) 2016
University of L'Aquila, University of Hamburg, Gdańsk University of Technology *Europe*
Joint Erasmus Mundus programme; thesis at the University of Stuttgart.

BE, Mechanical Engineering (with Distinction) 2012
University of Pune *Pune, India*

Teaching and mentoring

- Co-delivered **Machine Learning II** lectures and exercises (winter 2025/26), led **Information Geometry** exercises (summer 2026), and previously taught exercises for Machine Learning I and II.
- Organized exercises for **Optimal and Robust Control** (2018-2023), **System Identification** (2018, 2019, 2022), and **Control Systems Theory and Design** (2018); produced reusable and recorded teaching material.
- Supervised **20 bachelor's and master's students**; six projects led to peer-reviewed publications in machine learning, data-driven control, and robotics.
- Delivered an industry tutorial on integral quadratic constraints for IAV GmbH.

Selected research outputs

1. **V. Stein, A. Datar, and N. Ay.** "Well-Posed KL-Regularized Control via Wasserstein and Kalman-Wasserstein KL Divergences." *Proceedings of the 43rd International Conference on Machine Learning*, PMLR 306:115711-115746, 2026. [link]
2. **A. Datar and N. Ay.** "Information Geometry for Wasserstein KL Divergence of Gaussian Measures on $\mathbb{R}^n$." Submitted to *Information Geometry*; arXiv:2503.24022, v3, 2026. [link]
3. **A. Datar and N. Ay.** "Wasserstein KL-Divergence for Gaussian Distributions." *Geometric Science of Information*, 2025, pp. 91-101. [link]
4. **A. Datar and N. Ay.** "Convergence Properties of Natural Gradient Descent for Minimizing KL Divergence." *TMLR*, 2025. [link]
5. **N. Ay, J. van Oostrum, and A. Datar.** "On the Natural Gradient of the Evidence Lower Bound." *JMLR* 26, 2025. [link]
6. **C. Datar, A. Datar, F. Dietrich, and W. Schilders.** "Systematic Construction of Continuous-Time Neural Networks for Linear Dynamical Systems." *SIAM Journal on Scientific Computing* 47(4), 2025. [link]
7. **A. Datar, C. Hespe, and H. Werner.** "Robust Performance Analysis of Cooperative Control Dynamics via Integral Quadratic Constraints." *IEEE Transactions on Automatic Control* 70(1), 2025. [link]
8. **A. Datar, P. Paulsen, and H. Werner.** "Flocking Towards the Source: Indoor Experiments with Quadrotors." *European Control Conference*, 2020, pp. 1638-1643. [link]

Communication and open research

- Presented natural-gradient research at FDIG 2025, University of Tokyo, and at TU Hamburg's Mathematics of Machine Learning 2025 conference.
- Contributed to WiMAS, open-source wireless multi-agent simulation software, and produced public course recordings and experimental robotics demonstrations.

Methods and tools

Research: machine learning, information geometry, natural-gradient methods, optimization, reinforcement learning, system identification, robust control, multi-agent systems, and data-driven control.

Tools: Python, MATLAB, LaTeX, Claude Code, OpenAI Codex.

Languages: English (C1), German (B1), Marathi (native).